

NBR-RELATED PATENT LANDSCAPE REPORT BASED ON US10266615B2

1. ABSTRACT

This report synthesizes the supplied patent evidence for NBR-related chemistry, using the single manifest patent US10266615B2 as the source of technical detail. Although the patent is formally directed to an acrylonitrile-butadiene-styrene (ABS) graft copolymer rather than a standalone nitrile-butadiene rubber (NBR) product, it is relevant because it describes preparation of a diene-based rubber latex that serves as the rubber phase in subsequent grafting. The disclosed approach focuses on improving impact strength and processability by introducing an acrylate-based cross-linking agent during latex formation and then reusing the same cross-linker in the graft copolymerization step. The patent frames the problem as a tradeoff between higher rubber content and reduced graft efficiency, gloss, and thermal stability, while also noting that conventional latex preparation can require long reaction times and may suffer from reduced conversion above about 90% and from coagulation under harsher conditions.

The key synthesis sequence is a two-step emulsion process. In Step 1, a diene-based rubber latex is prepared from 50 to 75 parts by weight of conjugated diene monomer, with 1 to 3 parts emulsifier, 0.1 to 0.4 parts initiator, 0.1 to 3 parts electrolyte, 0.1 to 0.5 part molecular weight modifier, and 65 to 100 parts water, followed by addition of 0.05 to 0.3 parts cross-linker per 100 parts diene monomer. In Step 2, the same cross-linker is combined with an aromatic vinyl compound and a vinyl cyan compound for graft copolymerization. The exemplified cross-linker is propylene glycol diacrylate, with n ranging from 1 to 15 in the claim language. Reported process data include maturation for 1 hour, conversion reaching 60% and then 95% after additional butadiene feed and heating to 80 °C. Overall, the evidence supports a formulation strategy centered on controlled cross-linking, emulsion polymerization, and staged monomer addition to improve rubber-phase performance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US10266615B2
- Patent Title: Method for preparing acrylonitrile-butadiene-styrene graft copolymer and ...
- Assignee: Not disclosed

- Jurisdiction: US
- Publication Year: 2019-04-23
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Maturation: 1 hour (1 hour of Maturation) - Conversion: 60 % (conversion ... 60 %) - Conversion: 95 % (conversion ... 95 %) - Total: 100 parts by weight (100 parts by weight of Total) - When: 82 parts by weight (82 parts by weight of When) - When: 82 parts by weight (82 parts by weight of When)

Relevant Experimental Evidence - Example EXAMPLE: Conversion: 60 %, Conversion: 95 %, Total: 100 parts by weight, Maturation: 1 hour - Example COMPARATIVE EXAMPLE 1: - Example EXAMPLE 3: When: 82 parts by weight - Example EXAMPLE 4: When: 82 parts by weight - Example EXAMPLE 2: - Example EXAMPLE 2: - Example EXAMPLE 1: - Example EXAMPLE 4: - Example EXAMPLE 1:

Technical Relevance Selected via deterministic pipeline scoring.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

Based on the supplied evidence, the most defensible synthesis guidance is to use a staged emulsion-polymerization route for diene-based rubber latex, with controlled incorporation of an acrylate cross-linker such as propylene glycol diacrylate. The patent indicates useful operating windows of 50 to 75 parts diene monomer, 1 to 3 parts emulsifier, 0.1 to 0.4 parts initiator, 0.1 to 3 parts electrolyte, 0.1 to 0.5 part molecular weight modifier, and 65 to 100 parts water, followed by 0.05 to 0.3 parts cross-linker per 100 parts diene monomer. A maturation period of about 1 hour and staged heating to 80 °C were associated with conversion progression from 60% to 95% in the example. The main technical implication is that cross-linking during latex formation, combined with reuse of the same cross-linker during grafting, is intended to improve impact strength while maintaining processability. Because only one patent was provided, no broader patent-to-patent comparison can be made from the available record.

5. REFERENCES

- US10266615B2 | Method for preparing acrylonitrile-butadiene-styrene graft copolymer and ... | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10266615B2/en>